EE2001 Electrical Circuits- I:

Inductance, Capacitance and Mutual Inductance

27 November 2023

Outline



1. **The Inductor**
2. **The Capacitor**
3. **Series-Parallel Combinations of Inductance and Capacitance**

The Inductor

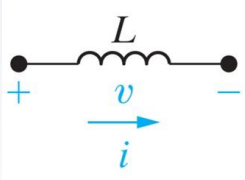
- An **inductor** is an **ideal circuit element**.
- It has the capacity to **store energy** (in its **magnetic field**). It **opposes any change in electrical current**. The source of the magneticfield is charge in motion, or **current**.
- It is composed of **a coil of wire wound around a supporting core, whose material may be magnetic or nonmagnetic.**
- However, it **cannot generate electric energy**. Therefore it is **a passive circuit element**.
- An inductor is described by its **inductance**.
- In circuits, inductance is usually denoted by a symbol *L* and has the SI unit Henry (H).
- The circuit symbol of an inductor is:



The Inductor

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- The **current and voltage characteristics** of an inductor can be expressed as follows:



$$v = L\frac{di}{dt}$$

- Under **passive sign convention**, this relation holds for the above voltage reference polarity and current direction (i.e., the reference direction of the current is **in the direction of reference voltage drop**).
- The voltage across the terminals of an inductor is proportional to the time rate of change of the current in the inductor.
- If the reference current enters from the negative terminal of the reference voltage (i.e., the reference direction of the current is in the direction of reference voltage rise) the current voltage equation becomes:

$$v = -L\frac{di}{dt}$$

The Inductor

- Note that, these is voltage across an inductor, only if there is change in the current.

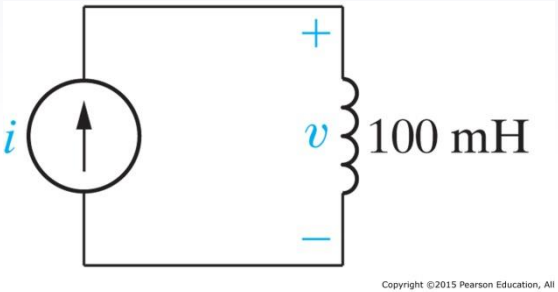
$$v = L\frac{di}{dt}$$

- If the current is varying with time, the magnetic field is varying with time. A varying magnetic field induces a voltage in any conductor linked by the field.
- The circuit parameter of inductance relates the induced voltage to the current.
- If the current through an inductor is **constant or DC current**, there is no voltage across it (i.e., the voltage is zero).
    - In the presence of a **constant current (DC), an inductor behaves as a short circuit**.
- **Current cannot change instantaneously in an inductor**. In switching, this usually **causes arcing**. After openning a switch, the currrent **initially continues to flow in the air across the switch**.
- Switching inducive circuits in an **important engineering problem**.

The Inductor

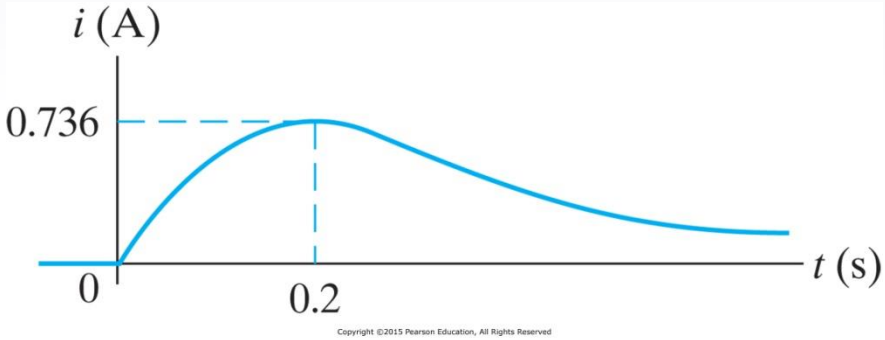
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- Example-1: The independent current source in the circuit is shown in the figure generates zero current for $t < 0$ and a pulse $10te^{-5t}\ A$ for $t > 0$.



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a) Skecth the current waveform.



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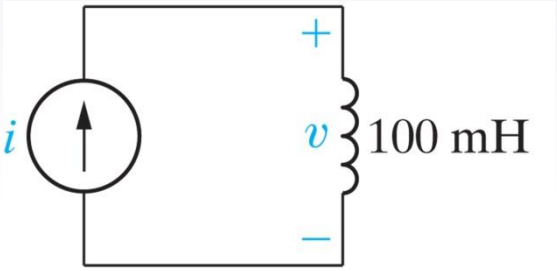
b) At what instant of time is the current maximum?

$$\frac{di}{dt} = 10(-5te^{-5t} + e^{-5t}) = 10(1 - 5t)e^{-5t}\ A/s$$

$$\frac{di}{dt} = 0 \quad \text{when } t = \frac{1}{5}\,\text{s}$$

The Inductor

- Example-1: The independent current source in the circuit is shown in Fig. 6.2 generates zero current for $t < 0$ and a pulse $10te^{-5t}\ A$ for $t > 0$.



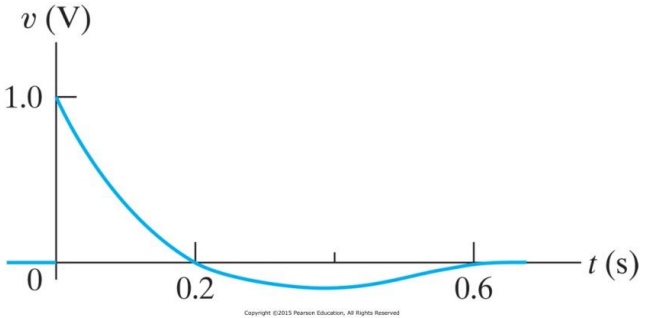
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c) Express the voltage across the terminals of the 100 mH inductor as a function of time.

$$v = L\frac{di}{dt} = (0.1)10e^{-5t}(1-5t) = e^{-5t}(1-5t)\ V\ ,\ t > 0$$

$$v = 0\ ,\ t < 0$$

d)Sketch the voltage waveform.

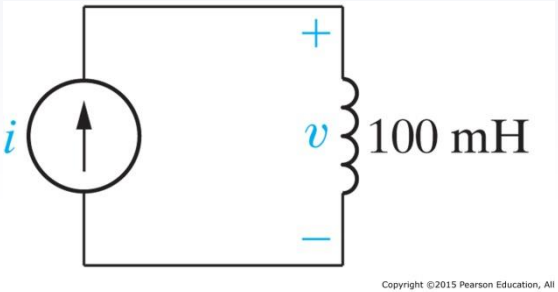


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The Inductor

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- Example-1: The independent current source in the circuit is shown in Fig. 6.2 generates zero current for $t < 0$ and a pulse $10te^{-5t}\ A$ for $t > 0$.



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e) Are the voltage and the current at a maximum at the same time?

No, the voltage is proportional to di/dt, not i.

f) At what instant of time does the voltage change polarity?

At 0.2 s, which corresponds to the moment when di/dt is passing through zero and changing sign.

g) Is there ever an instantaneous change in voltage across the inductor? If so, at what time?

Yes, at t=0. Note that the voltage can change instantaneously across the terminals of an inductor.

The Inductor

- We express the voltage across an inductor in terms of the current through it.
- It is also possible to express the current through an inductor in terms of the voltage across it.

$$i(t) = i(t_0) + \frac{1}{L}\int_{t_0}^{t} v(\tau)d\tau$$

- For $t_0 = 0$ this becomes

$$i(t) = i(0) + \frac{1}{L}\int_{0}^{t} v(\tau)d\tau$$

The Inductor

$$i(t) = i(t_0) + \frac{1}{L}\int_{t_0}^{t} v(\tau)d\tau$$

- Since the current is the integral of the voltage, it is a continuous variable.
  - It cannot have discontinuities (i.e., jump instantaneously) unless special circumstances.
- Note that

$$\int_{t_0}^{t} v(\tau)d\tau$$

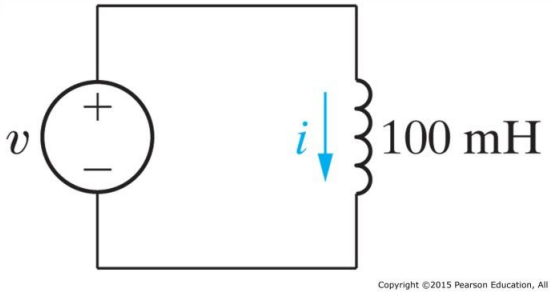
  is the area under the voltage $v(t)$ curve. The chage in the current is proportional to the area under the $v(t)$ curve.
- Note also that the above formulas hold under the condition that the reference direction of the current is in the direction of reference voltage drop. Otherwise, one needs to insert a minus sign into the formula.

The Inductor

- Example-2: The voltage pulse applied to the 100mH inductor is 0 for $t < 0$ and is given by the expression

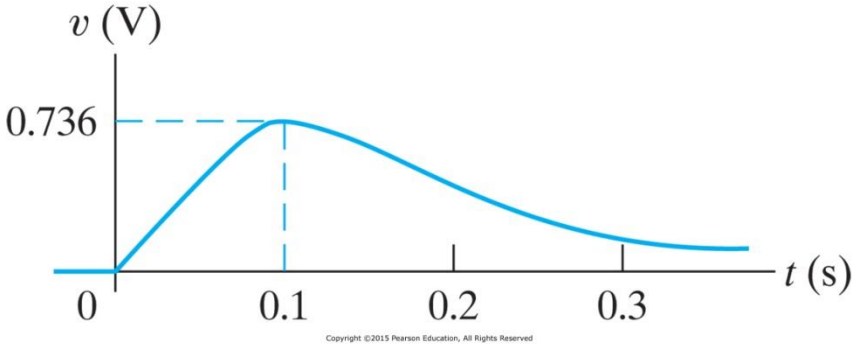
$$v(t) = 20te^{-10t}\ V$$

for $t > 0$. Also assume $i = 0$ for $t \leq 0$.



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a) Skecth the voltage as a function of time.



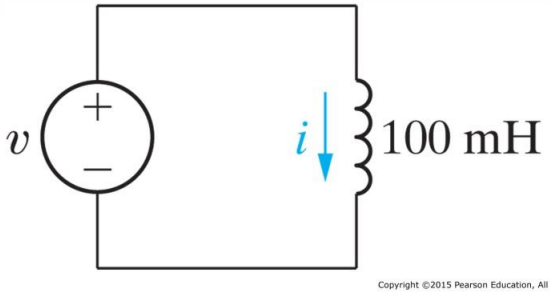
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The Inductor

- Example-2: The voltage pulse applied to the 100mH inductor is 0 for $t < 0$ and is given by the expression

$$v(t) = 20te^{-10t}\ V$$

for $t > 0$. Also assume $i = 0$ for $t \leq 0$.



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b) Find the inductor current as a function of time.

The current in the inductor is 0 at $t = 0$. Therefore, the current for $t > 0$ is

$$i = \frac{1}{0.1}\int_0^t 20\tau\, e^{-10\tau} d\tau + 0$$

$$= 200\left[e^{-10\tau}\left(\frac{-10\tau - 1}{100}\right)\right]\Bigg|_0^t$$

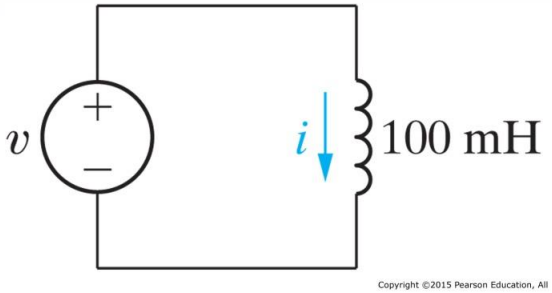
$$= 2(1 - 10te^{-10\tau} - e^{-10\tau})\ A, \quad t > 0$$

The Inductor

- Example-2: The voltage pulse applied to the 100mH inductor is 0 for $t < 0$ and is given by the expression

$$v(t) = 20te^{-10t}\ V$$

for $t > 0$. Also assume $i = 0$ for $t \leq 0$.

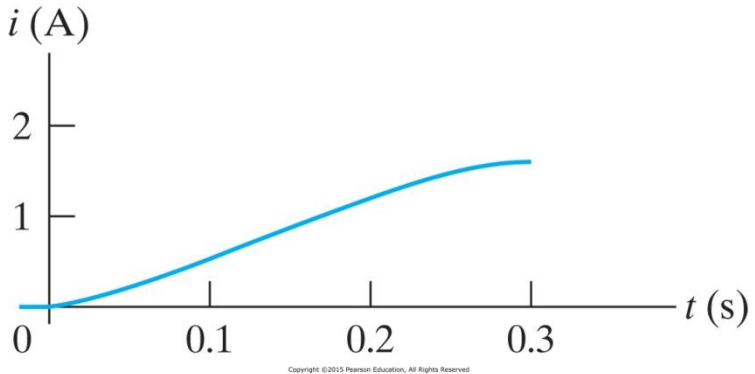


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c) Skecth the current as a function of time.

$$v = 0, \qquad t < 0$$

$$v = 20te^{-10t}\ V, \qquad t > 0$$



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Power and Energy in the Inductor

- Given the current-voltage characteristics of the inductor one can easily derive its power and energy relationships.

$$p = vi \qquad p = Li\frac{di}{dt} \qquad p = v\left[i(t_0) + \frac{1}{L}\int_{t_0}^{t} v(\tau)d\tau\right]$$

- To derive the energy one can use the equation

$$p = \frac{dw}{dt} = Li\frac{di}{dt}$$

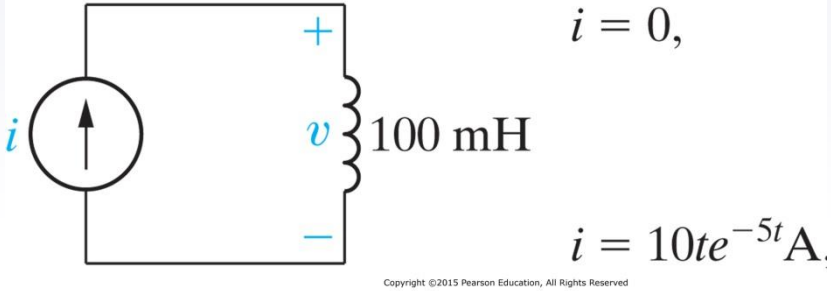
- Rearranging, integrating both sides, and under the assumption that $i = 0$ results in $w = 0$, one obtains

$$w = \frac{1}{2}Li^2$$

The Inductor

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- Example-3: Consider the circuit with the current behavior given below.



$$i = 0,$$

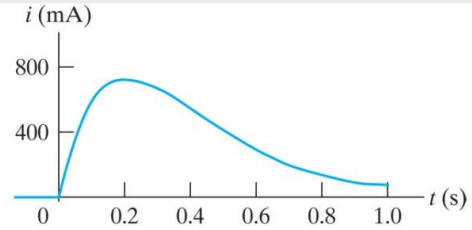
$$i = 10te^{-5t}\text{A}$$

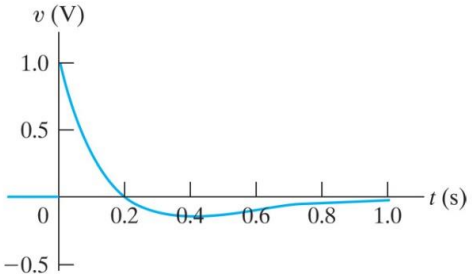
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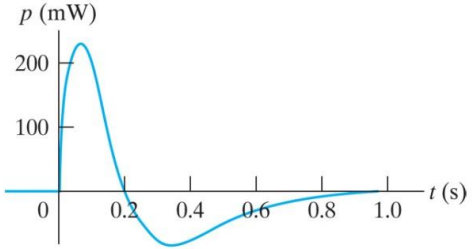
a) Plot i, v, p and w versus time. Line up the plots verticaly to allow easy assessment of each variable's behavior.

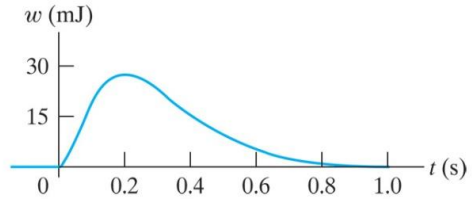
Based on the i and v obtained in Example-1 and considering $p = vi$ and

$w = \left(\frac{1}{2}\right) Li^2,$







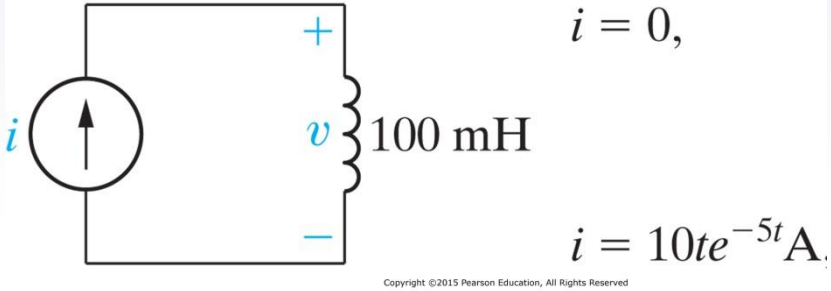


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The Inductor

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- Example-3: Consider the circuit with the current behavior given below.



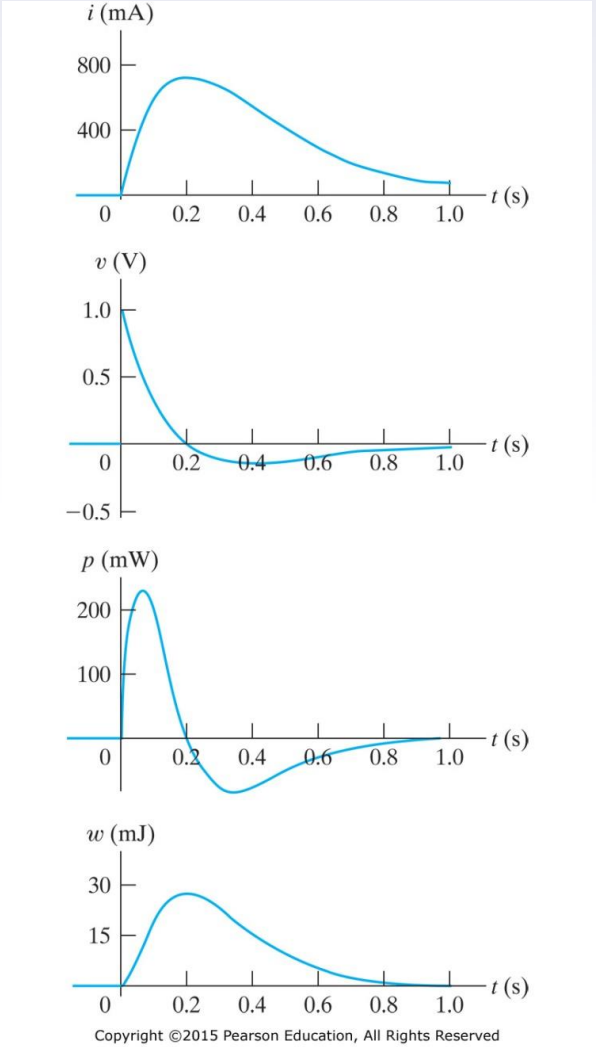
$i = 0,$

$i = 10te^{-5t}\text{A}$

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b) In what time interval is energy being stored in the inductor?

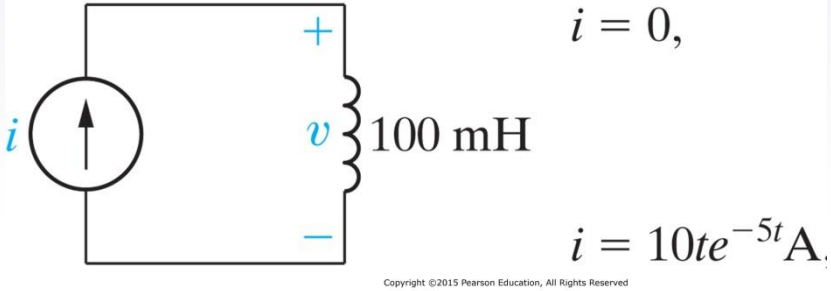
An increasing energy curve indicates that energy is being stored. Thus energy is being stored in the time interal 0 to 0.2 s. Note that this corresponds to the interval when $p > 0$.



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The Inductor

- Example-3: Consider the circuit with the current behavior given below.



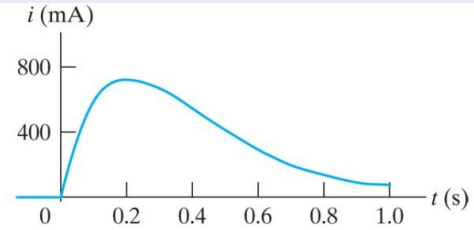
$i = 0,$

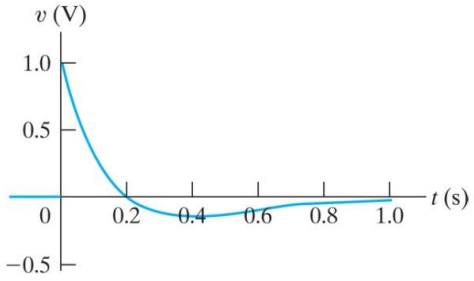
$i = 10te^{-5t}\text{A}$

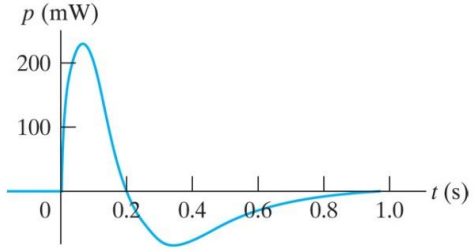
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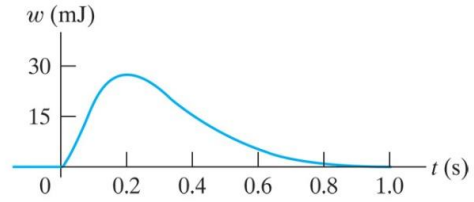
c) In what time interval is energy being extracted from the inductor?

A decreasing energy curve indicates that energy is being extracted. Thus energy is being extracted in the time interal 0.2 s to ∞. Note that this corresponds to the interval when $p < 0$.





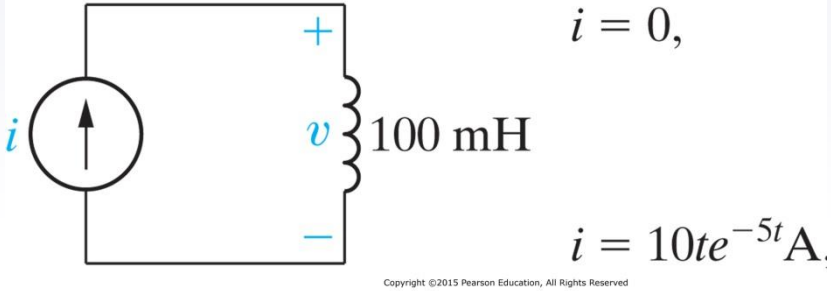




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The Inductor

- Example-3: Consider the circuit with the current behavior given below.



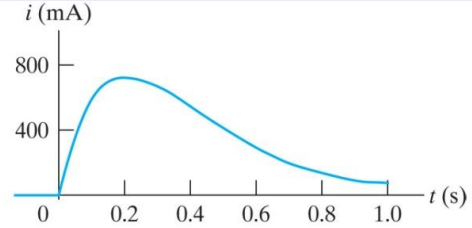
$i = 0,$

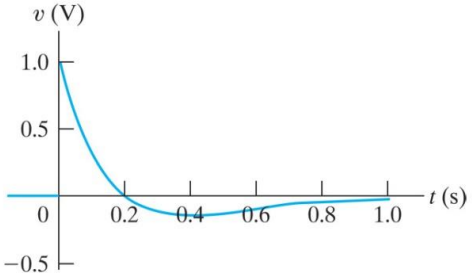
$i = 10te^{-5t}\text{A}$

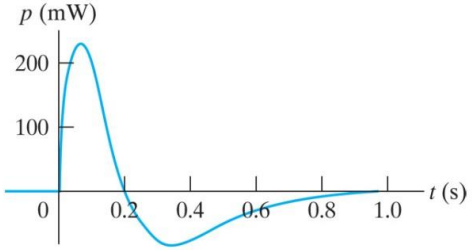
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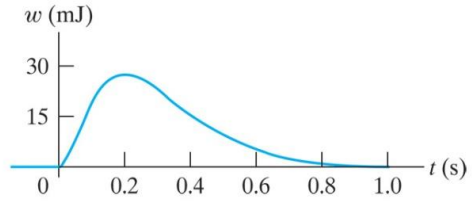
d) What is the maximum energy stored in the inductor?

We see that energy is at a maximum when current is at a maximum. The graphs also confirm this. In Example-1, maximum current was calculated as 0.736 A. Therefore $w_{max} = 27.07\ mJ$.





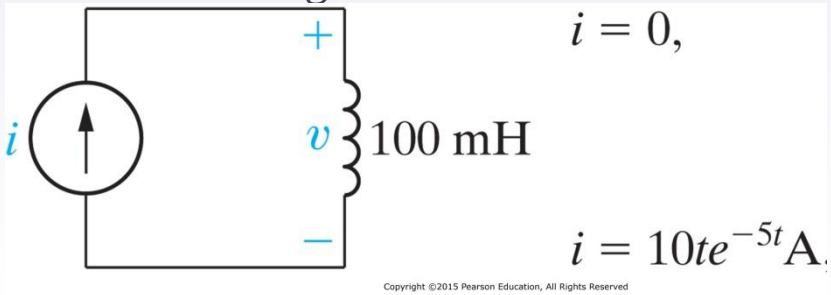




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The Inductor

- Example-3: Consider the circuit with the current behavior given below.



$i = 0,$

$i = 10te^{-5t}\text{A}$

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e) Evaluate the integrals

$$\int_0^{0.2} p\,dt \qquad \int_{0.2}^{\infty} p\,dt$$

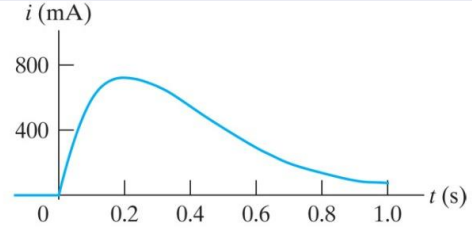
and comment on their significance.

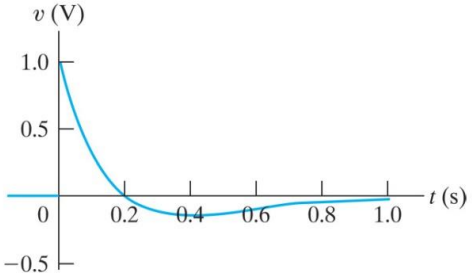
From Example-1,

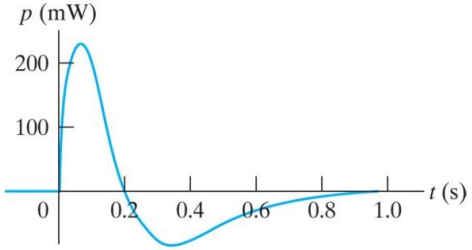
$i = 10te^{-5t}\ A$ and $v = e^{-5t}(1 - 5t)\ V$

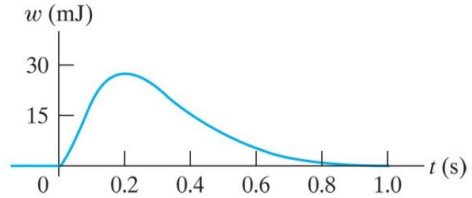
Therefore,

$p = vi = 10te^{-10t} - 50t^2e^{-10t}\ W$





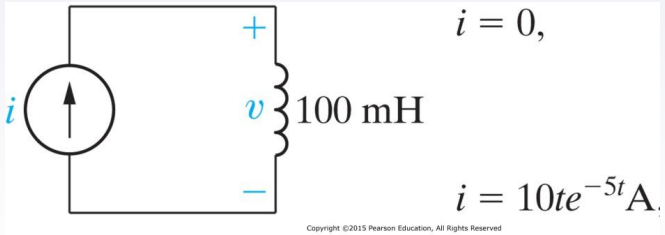




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The Inductor

- Example-3: Consider the circuit with the current behavior given below.



$i = 0,$

$i = 10te^{-5t}\text{A}$

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Continuation of e):

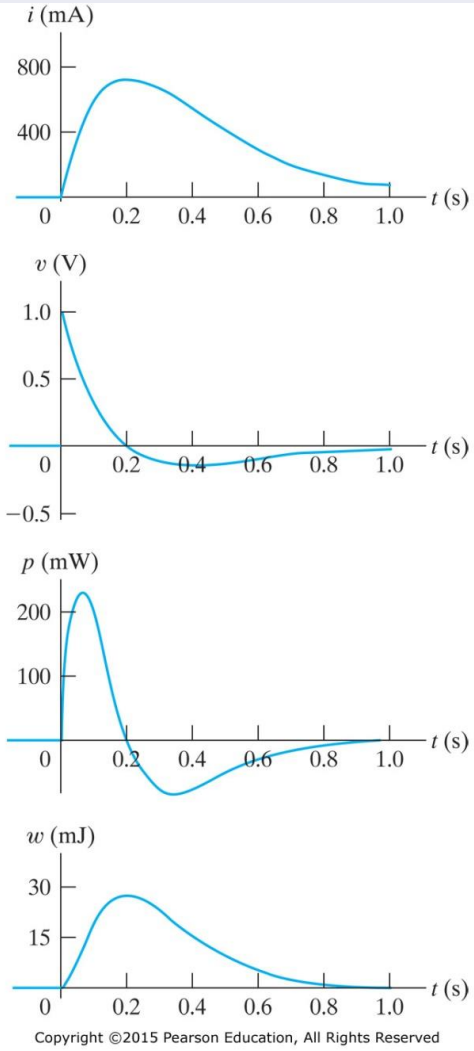
$$\int_0^{0.2} p\,dt = 10\left[e^{-10t}\left(\frac{-10t-1}{100}\right)\right]\Bigg|_0^{0.2}$$

$$-50\left\{\frac{t^2e^{-10t}}{-10} + \frac{2}{10}\left[e^{-10t}\left(\frac{-10t-1}{100}\right)\right]\right\}\Bigg|_0^{0.2}$$

$$= 0.2e^{-2} = 27.07\ mJ$$

$$\int_{0.2}^{\infty} p\,dt = 10\left[e^{-10t}\left(\frac{-10t-1}{100}\right)\right]\Bigg|_{0.2}^{\infty}$$

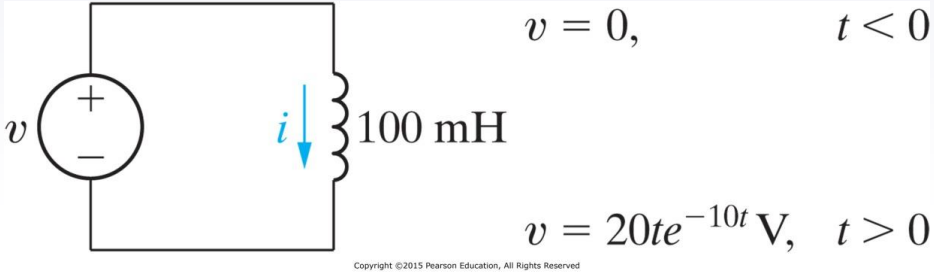
$$-50\left\{\frac{t^2e^{-10t}}{-10} + \frac{2}{10}\left[e^{-10t}\left(\frac{-10t-1}{100}\right)\right]\right\}\Bigg|_{0.2}^{\infty} = -0.2\,e^{-2} = -27.07\ mJ$$



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The Inductor

- Example-3: Consider the circuit with the current behavior given below.



$$v = 0, \qquad t < 0$$

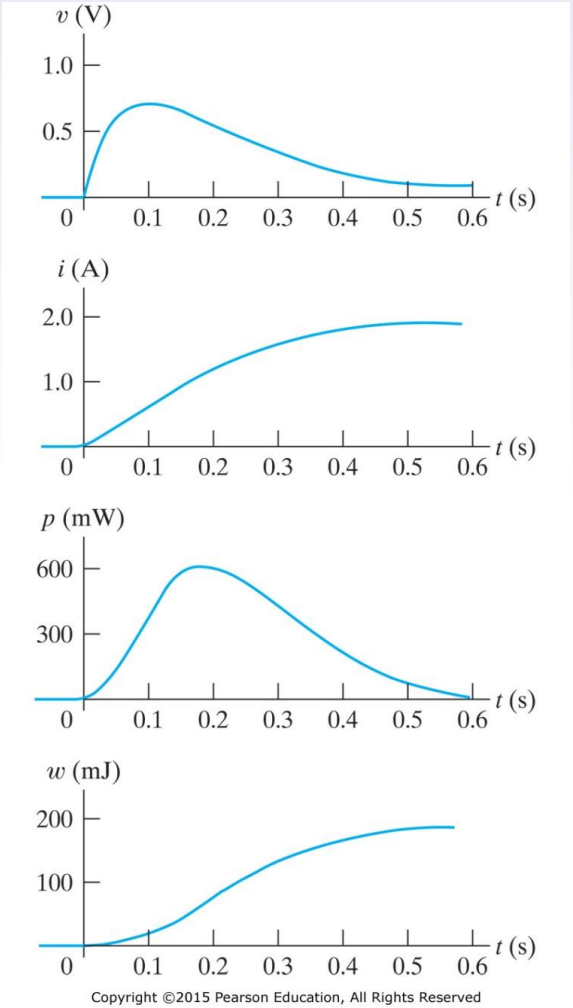
$$v = 20te^{-10t}\ \text{V}, \qquad t > 0$$

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f) Considering the circuit of Example-2, plot i, v, p and w versus time. Line up the plots verticaly to allow easy assessment of each variable's behavior.

Based on the i and v given in Example-2 and considering $p = vi$ and

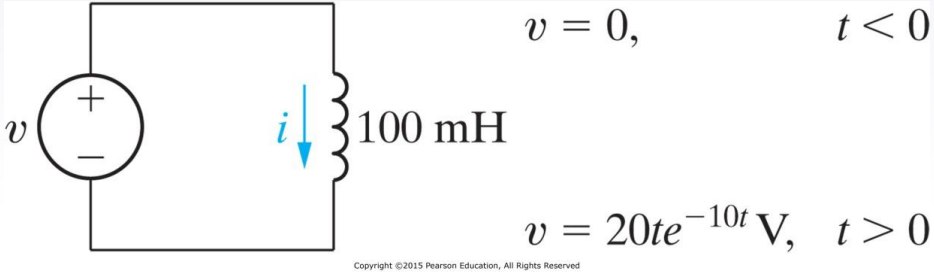
$w = \left(\frac{1}{2}\right) Li^2$,



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The Inductor

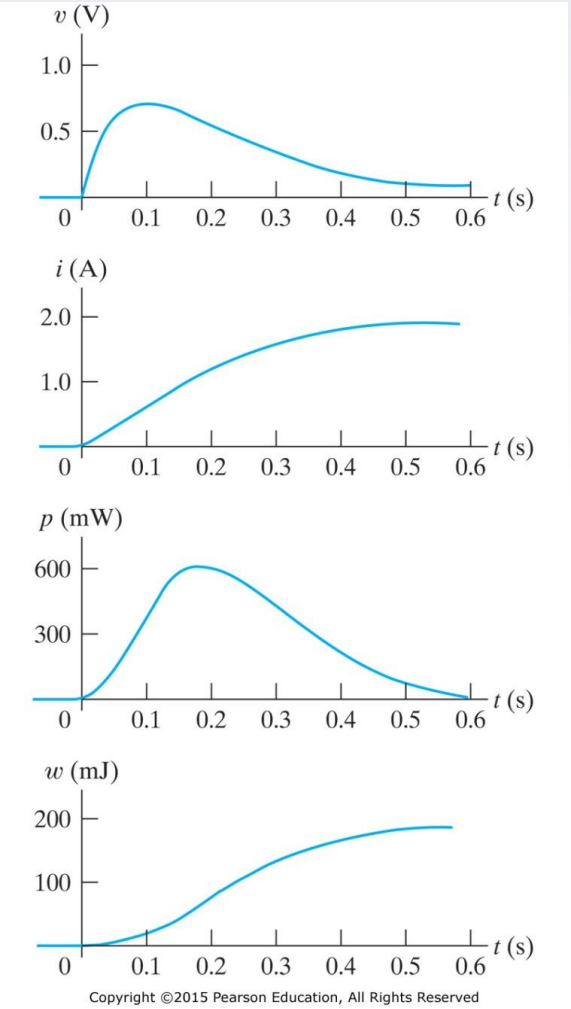
- Example-3: Consider the circuit with the current behavior given below.



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g) In what time interval is energy being stored in the inductor?

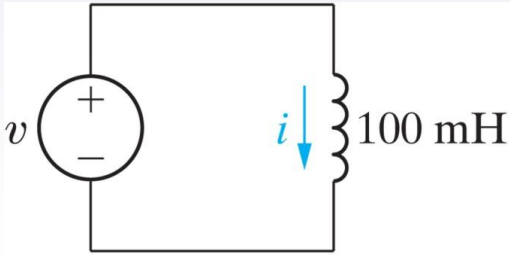
The power is always positive, and hence energy is always being stored during the voltage pulse.



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The Inductor

- Example-3: Consider the circuit with the current behavior given below.



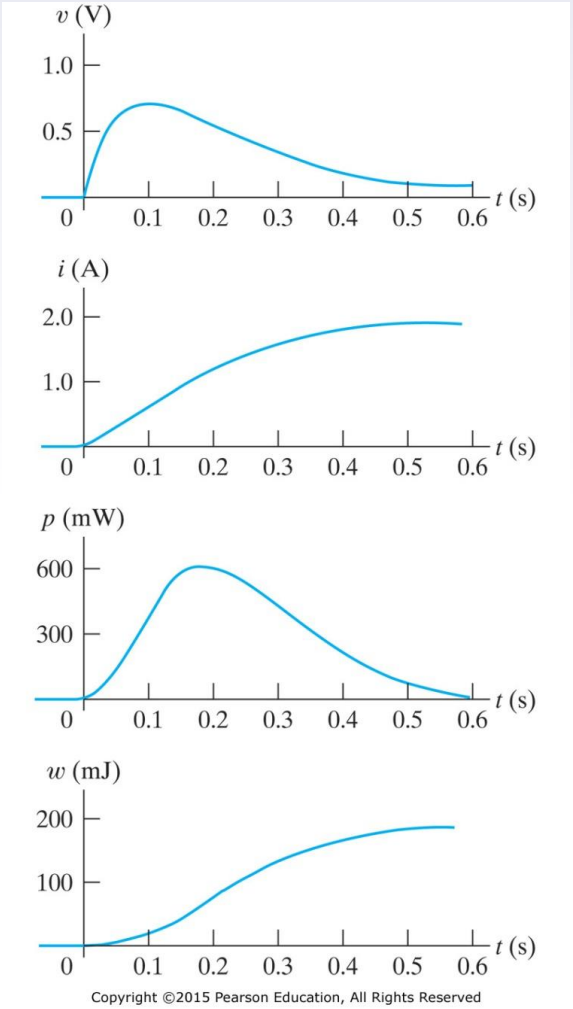
$$v = 0, \qquad t < 0$$

$$v = 20te^{-10t}\ \text{V}, \qquad t > 0$$

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h) In Example-2, why is there a sustained current in the inductor as the voltage approaches zero?

The application of the voltage pulse stores energy in the inductor. Because the inductor is ideal, this energy cannot dissipate after the voltage subsides to zero. Therefore, a sustained current circulates in the circuit. A lossless inductor is obviously an ideal circuit element. Practical inductors require a resistor in the circuit model (will be discussed later).



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The Capacitor

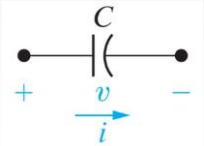
2

- A capacitor is another ideal circuit element.
- It is an electrical component that consists of two conductors seperated by an insulator or dielectric material.
- It has the capacity to store energy (in its electric field). The source of the electric field is seperation of charge, or voltage.
- However, it cannot generate electric energy. Therefore it is a passive circuit element.
- A capacitor is described by its capacitance.
- In circuits, capcitance is usually denoted by a symbol C and has the SI unit Farad (F).
- Farad is a very large quantity. Usually we use mF, μF, nf and pF.
- The circuit symbol of a capacitance is:



2 The Capacitor

- The current and voltage characteristics of a capacitor can be expressed as follows:



$$i = C\frac{dv}{dt}$$

- Under passive sign convention, this relation holds for the above voltage reference polarity and current direction (i.e., the reference direction of the current is in the direction of reference voltage drop).
- If the reference current enters from the negative terminal of the reference voltage (i.e., the reference direction of the current is in the direction of reference voltage rise) the current voltage equation becomes:

$$i = -C\frac{dv}{dt}$$

The Capacitor

2

- The current through a capacitor depends on the change in the voltage across its terminals

$$i = C\frac{dv}{dt}$$

- If the voltage is varying with time, the electric field is varying with time. A varying electric field produces a displacement current in the space occupied by the field.
- The circuit parameter of capacitance relates the displacement current to the voltage, where the displacement current is equal to the conduction current at the terminals of the capacitor.
- If the voltage across a capacitor is constant, there is no current through it (i.e., the current is zero).
    - In the presence of a constant voltage, a capacitor behaves as a open circuit.

The Capacitor

- We express the current through a inductor in terms of the voltage across it.
- It is also possible to express the voltage across a capacitor in terms of the current through it.

$$v(t) = v(t_0) + \frac{1}{C}\int_{t_0}^{t} i(\tau)d\tau$$

- For $t_0 = 0$ this becomes

$$v(t) = v(0) + \frac{1}{C}\int_{0}^{t} i(\tau)d\tau$$

The Capacitor

$$v(t) = v(t_0) + \frac{1}{C}\int_{t_0}^{t} i(\tau)d\tau$$

- Since the voltageis the integral of the current, it is a continuous variable.
  - It cannot have discontinuities (i.e., jump instantaneously) unless special circumstances.
- Note that

$$\int_{t_0}^{t} i(\tau)d\tau$$

  is the area under the voltage i(t) curve. The chage in the voltage is proportional to the area under the $i(t)$ curve.
- Note also that the above formulas hold under the condition that the reference direction of the current is in the direction of reference voltage drop. Otherwise, one needs to insert a minus sign into the formula.

Power and Energy in the Capacitor

- Given the current-voltage characteristics of the capacitor one can easily derive its power and energy relationships.

$$p = vi \qquad p = Cv\frac{dv}{dt} \qquad p = i\left[v(t_0) + \frac{1}{C}\int_{t_0}^{t} i(\tau)d\tau\right]$$

- To derive the energy one can use the equation

$$p = \frac{dw}{dt} = Cv\frac{dv}{dt}$$

- Rearranging, integrating both sides, and under the assumption that $i = 0$ results in $w = 0$, one obtains

$$w = \frac{1}{2}Cv^2$$

Summary

TABLE 6.1 Terminal Equations for Ideal Inductors and Capacitors

Inductors	
$v = L\frac{di}{dt}$	(V)
$i = \frac{1}{L}\int_{t_0}^{t} v\,d\tau + i(t_0)$	(A)
$p = vi = Li\frac{di}{dt}$	(W)
$w = \frac{1}{2}Li^2$	(J)
Capacitors	
$v = \frac{1}{C}\int_{t_0}^{t} i\,d\tau + v(t_0)$	(V)
$i = C\frac{dv}{dt}$	(A)
$p = vi = Cv\frac{dv}{dt}$	(W)
$w = \frac{1}{2}Cv^2$	(J)

TABLE 6.2 Equations for Series- and Parallel-Connected Inductors and Capacitors

Series-Connected
$L_{eq} = L_1 + L_2 + \cdots + L_n$
$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots + \frac{1}{C_n}$
Parallel-Connected
$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \cdots + \frac{1}{L_n}$
$C_{eq} = C_1 + C_2 + \cdots + C_n$

End of the Lesson (27.11.23)

Any questions?

Supplementary videos: Enter Youtube-> Search for ‘Alparslan Zehir’ -> Go to Playlists -> Watch ‘Electrical Circuits’ playlist